# Eva: Evidence-Preserving Temporal Window Compression for Stride-Robust Video Action Recognition


## Abstract

Temporal stride sampling is the default efficiency primitive in video action recognition, but it is a lossy measurement process: if a stride grid skips the few frames containing handshape changes, contacts, object-state transitions, or phase boundaries, the recogniser receives a distorted observation rather than a cheaper version of the same evidence. Prior large-scale analysis across 8 architectures and 8 benchmarks documents this failure mode: FineGym and AUTSL reach architecture-averaged dense-to-stride-16 gaps of 55.9 pp and 53.0 pp (*Temporal Demand Score*), while confidence concentration predicts per-class aliasing loss across all benchmarks. We propose **Eva** (**E**vidence-preserving **V**ideo **A**nti-aliasing), a pre-decimation window-compression module for stride-robust recognition. Unlike frame-selection methods, Eva observes every frame in a stride window before reducing temporal resolution; unlike classical low-pass anti-aliasing, it explicitly preserves high-frequency discriminative residuals instead of suppressing them. Each window is represented by low-frequency context and event-residual tokens, trained with dense-to-sparse consistency, evidence alignment, and sampling-phase robustness. We evaluate Eva as a backbone-agnostic front-end and instantiate its strongest variant with a bidirectional Mamba-3 temporal backbone. Experiments target high-temporal-demand benchmarks (FineGym, AUTSL, SSv2) and low-demand controls (UCF-101, HMDB-51) [1, 2, 3, 4, 5], reporting accuracy across $s \in \{1, 2, 4, 8, 16\}$, TDS reduction, Anti-Aliasing Gain, phase robustness, and end-to-end accuracy–latency Pareto curves.


## 1 Motivation

Standard sparse sampling with stride $s$ constructs $X^{(s)}_{\text{stride}} = \{x_1, x_{1+s}, x_{1+2s}, \ldots\}$, discarding all intermediate frames. This is benign only when class evidence is redundant across the omitted frames. If a discriminative event occupies $d$ frames inside a stride window of length $s$, a one-frame sampler with random phase observes that event with probability at most $\min(1, d/s)$. More generally, sampling $k$ frames without replacement misses the event with probability

$$\frac{\binom{s-d}{k}}{\binom{s}{k}},$$

so high-stride inference is inherently phase-sensitive when $d \ll s$.

Systematic empirical analysis [6] across 8 architectures, including 3D CNNs, efficient temporal modules, video transformers, and SSMs [7, 8, 9, 10, 11, 12], and 8 benchmarks reveals two key findings. First, accuracy degradation under aggressive stride is *structured*: datasets whose identity hinges on brief events (FineGym, TDS=55.9 pp; AUTSL, TDS=53.0 pp; SSv2, TDS=27.3 pp) suffer consistently across all architectures, while appearance-dominated datasets are nearly unaffected (UCF-101, TDS=4.9 pp). Second, whole-frame optical-flow frequency is an *anti*-correlated proxy for vulnerability ($\rho$=−0.64, $p$=$10^{-5}$, $n$=40): high-motion classes remain recognisable from any frame, while low-motion classes with brief key events are destroyed by the first dropped window. Instead, per-class *confidence concentration*—the temporal localisation of the classifier's correct-class softmax under a sliding window—predicts aliasing loss positively ($\rho$=+0.26 pooled, positive in all 8 datasets). This agrees with prior temporal-dataset studies showing that some action categories require temporal order while others are solvable from static appearance [13, 14].

**Hypothesis.** *Sparse recognition should not be implemented as frame dropping. Before reducing temporal resolution, the model should compress each stride window into an evidence-preserving representation.*

## 2 Positioning

Eva is positioned against four related lines of work:

**Sparse temporal recipes.** Segment-based sampling and temporal shift modules made sparse video inference practical [15, 16], but they still assume that a small set of sampled frames is a sufficient statistic for the clip. Our premise is different: high-TDS classes contain windows where the unsampled frames carry the only class-discriminative evidence.

**Adaptive frame selection.** AdaFrame [17], MGSampler [18], SMART [19], FrameExit [20], AdaFocus [21], and MAFormer [22] adapt computation by selecting frames, clips, regions, or exits. They reduce cost, but selected observations are still a subset of the window. If the discriminative event falls in an unselected frame it is irreversibly lost. EVA compresses the full window before temporal decimation.

**Classical temporal anti-aliasing.** BlurPool [23] applies a low-pass filter before downsampling to recover shift-invariance in spatial CNNs. Temporal action-localisation work has likewise shown that low-pass filtering can mitigate downsampling artifacts, while warning that high-frequency bands contain task-critical information [24]. EVA follows the sampling-theoretic principle of pre-decimation filtering, but is not a pure low-pass filter: it preserves the event residual $r_t = f_t - \ell_t$ in a separate branch.

**Video SSMs.** VideoMamba [12] and related state-space architectures [25, 26, 27, 28] apply an efficient SSM to whatever token sequence is fed as input. They do not prevent evidence loss that occurs *before* the backbone: if the event frame was dropped, no SSM can recover it. EVA and Mamba-3 address orthogonal failure modes of sparse video inference: EVA protects evidence before sampling, while the backbone models the resulting compressed token sequence.

# 3 Core Idea

For stride $s$, the video is partitioned into windows $\mathcal{W}_i^{(s)} = \{x_{(i-1)s+1}, \ldots, x_{is}\}$. Instead of selecting one frame per window, EVA produces

$$z_i^{(s)} = \text{EVA}\left(\mathcal{W}_i^{(s)}\right),$$

yielding $Z^{(s)} = \{z_i^{(s)}\}_{i=1}^{\lceil T/s \rceil}$—same length as a stride-$s$ sample, but each token summarises the full window. The objective is therefore to preserve a sufficient statistic for class evidence under a fixed output-token budget:

$$I\left(Y; z_i^{(s)}\right) \gg I\left(Y; x_{(i-1)s+\phi}\right)$$

when evidence is localised inside $\mathcal{W}_i^{(s)}$. We do not claim that EVA reconstructs the dropped frames; it only preserves the discriminative evidence needed by the classifier.

The complete pipeline is:

$$X \xrightarrow{F} \{f_t\} \xrightarrow{\text{EVA}_s} Z^{(s)} \xrightarrow{\text{Mamba-3}} H^{(s)} \xrightarrow{\text{head}} p^{(s)},$$

where $F$ is a spatial encoder producing per-frame features $f_t \in \mathbb{R}^C$. For fair deployment claims we evaluate two settings: *feature-level EVA*, which isolates the compression effect using dense frame features, and *compute-aware EVA*, where latency includes decoding, spatial feature extraction, compression, and the temporal backbone. SOTA claims are made only on end-to-end accuracy–latency Pareto curves.

# 4 Architecture

## 4.1 EVA: Dual-Band Tokenizer

**Dual-band decomposition.** Given dense features $\{f_t\}$, EVA decomposes each frame into a smooth context signal and a high-frequency event residual:

$$\ell_t = \sum_{k=-K}^{K} h_k\, f_{t+k}, \qquad h_k \propto e^{-k^2/2\sigma_g^2}, \tag{1}$$

$$r_t = f_t - \ell_t, \tag{2}$$

where $\sigma_g > 0$ is a learnable bandwidth (init $\sigma_g$=1.5, $K$=3). $\ell_t$ captures slowly varying motion and appearance; $r_t$ encodes deviations from local mean—brief transient events.

**Event scoring.** A discriminative salience score identifies frames that carry class-specific evidence:

$$u_t = [f_t,\ |f_{t+1} - f_t|,\ |f_{t-1} - f_t|,\ |r_t|], \tag{3}$$

$$e_t = \text{sigmoid}(\text{MLP}_e(u_t)) \in [0, 1]. \tag{4}$$

The inputs encode the frame's absolute feature, its forward and backward temporal derivatives, and the magnitude of its event residual.

**Window compression.** For each window, EVA produces two complementary tokens:

$$z_{i,\text{ctx}}^{(s)} = \sum_{t \in \mathcal{W}_i} \alpha_{i,t}^{\text{ctx}}\, \ell_t, \tag{5}$$

$$z_{i,\text{evt}}^{(s)} = \sum_{t \in \mathcal{W}_i} \alpha_{i,t}^{\text{evt}}\, r_t, \tag{6}$$

where $\alpha^{\text{ctx}} = \text{softmax}(q^T \ell_t / \sqrt{C})$ with learned query $q$, and $\alpha^{\text{evt}} = \text{softmax}(e_t/\tau)$ with temperature $\tau$=0.1. If the most salient frame in a window exceeds all others by margin $m$, this softmax assigns it weight at least

$$\left(1 + (s-1)e^{-m/\tau}\right)^{-1},$$

so low temperature approximates event selection while remaining differentiable. The anti-aliased token is:

$$z_i^{(s)} = \mathrm{LN}\Big(W_{\mathrm{ctx}}\, z_{i,\mathrm{ctx}}^{(s)} + W_{\mathrm{evt}}\, z_{i,\mathrm{evt}}^{(s)} + p_i^{(s)}\Big)\,, \tag{7}$$

with $p_i^{(s)}$ a stride-conditioned learnable positional embedding. For large strides ($s \geq 8$), a two-token variant ($M$=2) exposes context and event as separate backbone tokens.

### 4.2 Backbone Instantiations

The primary implementation path uses TimeSformer [9], because it is a mature video-pretrained transformer with a well-established stride-sampling baseline. In **EVA-TimeSformer**, a frozen or lightly fine-tuned TimeSformer encoder maps dense frames to per-frame features, Eva compresses those features into stride-window tokens, and a temporal transformer head classifies the compressed sequence. The matched baseline is the same TimeSformer checkpoint evaluated with uniform stride sampling, so any improvement can be attributed to pre-decimation evidence compression rather than a new backbone.

We also test Mamba-2/VideoMamba and Mamba-3 [28, 29] as SSM instantiations. Mamba-3 is not the first experiment because its video weights are not yet established; it is an ablation for whether complex-valued selective states improve event tracking once Eva has already preserved the evidence. The paper's primary claim is therefore backbone-agnostic compression, with TimeSformer as the main SOTA-facing instantiation and Mamba-3 as an exploratory SSM variant.

## 5 Training Objective

We sample $s \sim \mathcal{U}\{1,2,4,8,16\}$ per step and optimise:

$$\begin{aligned}\mathcal{L} = \mathcal{L}_{\mathrm{cls}} + \lambda_{\mathrm{KL}}\mathcal{L}_{\mathrm{KL}} + \lambda_{\mathrm{evid}}\mathcal{L}_{\mathrm{evid}} \\ + \lambda_{\mathrm{phase}}\mathcal{L}_{\mathrm{phase}} + \lambda_{\mathrm{ent}}\mathcal{L}_{\mathrm{ent}}.\end{aligned} \tag{8}$$

$\mathcal{L}_{\mathrm{cls}}$: cross-entropy at all strides.

$\mathcal{L}_{\mathrm{KL}} = T_d^2\, \mathrm{D}_{\mathrm{KL}}\big(\mathrm{sg}(p^{(1)}) \| p^{(s)}\big)$: dense-to-sparse distillation; transfers calibrated confidence from the dense model to each sparse stride.

$\mathcal{L}_{\mathrm{evid}}$: evidence-map alignment. The dense pass produces a frame-level teacher map $a^{(1)}$ using either leave-one-frame-out confidence drop or gradient attribution over $f_t$:

$$a_t^{(1)} \propto p_y^{(1)}(X) - p_y^{(1)}(X_{\setminus t}) \quad \text{or} \quad a_t^{(1)} \propto \left\| \frac{\partial \log p_y^{(1)}}{\partial f_t} \right\|_1.$$

Sparse event weights are lifted back to the dense frame grid as $\hat{a}^{(s)}$, and

$$\mathcal{L}_{\mathrm{evid}} = \Big\| \mathrm{norm}(a^{(1)}) - \mathrm{norm}(\hat{a}^{(s)}) \Big\|_1.$$

This makes the event branch match dense evidence rather than merely discovering high-motion frames.

$\mathcal{L}_{\mathrm{phase}} = \mathrm{D}_{\mathrm{JS}}\big(p^{(s,\phi_1)} \| p^{(s,\phi_2)}\big)$: sampling-phase consistency; penalises predictions that vary with the temporal offset of the stride grid.

$\mathcal{L}_{\mathrm{ent}}$: target-entropy regularisation,

$$\mathcal{L}_{\mathrm{ent}} = \frac{1}{T/s} \sum_i \big(H(\alpha_i^{\mathrm{evt}}) - \log k\big)^2.$$

Here $k$ is the expected number of useful subevents per window. This avoids the failure mode of forcing a single peak when signs, contacts, or gymnastic phases require multiple frames.

## 6 Evaluation Plan

**Metrics.** Top-1 accuracy at each stride; $\mathrm{TDS}(m) = [a^{(1)} - a^{(16)}]_+$; Anti-Aliasing Gain

$$\mathrm{AAG} = \frac{1}{4} \sum_{s \in \{2,4,8,16\}} \Big(a_{\mathrm{EVA}}^{(s)} - a_{\mathrm{base}}^{(s)}\Big)\,;$$

phase robustness $\sigma^2_{\mathrm{phase}}$ across stride offsets; and accuracy–latency Pareto curves. Latency includes decoding, resize, spatial encoding, Eva, and the temporal backbone, measured at batch size 1 and 8.

**Datasets.** *High demand*: FineGym [1] (TDS=55.9 pp), AUTSL [2] (TDS=53.0 pp), SSv2 [3] (TDS=27.3 pp). *Medium*: Diving-48 [30], HMDB-51 [5]. *Low (control)*: UCF-101 [4] (TDS=4.9 pp).

**Baselines.** *Sampling*: uniform stride, TSN-style segment sampling [15], random phase, center-window, and dense upper bound. *Anti-aliasing*: temporal Gaussian low-pass, BlurPool-style filtering [23], and dynamic-cutoff LPF adapted from temporal action localisation [24]. *Adaptive inference*: AdaFrame [17], MGSampler [18], SMART [19], FrameExit [20], and AdaFocus [21]. *Token compression*: ToMe [31] and Eventful Transformers [32]. *Backbones*: TimeSformer [9], VideoMAE [11], ViViT [10], SlowFast [8], VideoMamba [12], Mamba-2 [28], and Mamba-3 [29]; each is tested with and without Eva.

## 7 Ablation Design

1. **Event branch vs. context-only**: ablate $z_{i,\mathrm{evt}}$=0; expected finding—loss concentrated on

FineGym, AUTSL, SSv2; negligible on UCF-101. Validates the dual-band design.

2. **Low-pass only vs. residual-preserving**: compare classical low-pass filtering to Eva's dual-band token. Validates that high-frequency residuals are useful rather than noise.

3. $\mathcal{L}_{\text{KL}}$: removal reduces sparse-stride accuracy without affecting $s$=1, confirming distillation as a stride-robustness mechanism independent of EVA.

4. $\mathcal{L}_{\text{phase}}$: removal increases $\sigma^2_{\text{phase}}$ on high-demand datasets; no effect on low-demand controls.

5. $\mathcal{L}_{\text{evid}}$: removal causes the event branch to diverge from dense evidence maps; measures the contribution of supervised evidence alignment.

6. **Entropy target** $k$: compare $k$=1 against multi-event settings; expected gain on sign language and fine-grained sports.

7. **Mamba-3 vs. Mamba-2 backbone**: complex states (Mamba-3) vs. real states (Mamba-2) with identical parameter count; expected gap largest on AUTSL (fine phoneme-level state tracking).

8. **Architecture generality**: Eva prepended to TimeSformer, VideoMAE, VideoMamba; EVA-TimeSformer is the primary SOTA-facing model, while EVA-Mamba3 tests the SSM variant.

9. **End-to-end compute fairness**: feature-level Eva vs. compute-aware Eva; gains must persist when spatial encoding cost is included.

# 8 Acceptance-Critical Results

The method should be considered successful only if it satisfies three empirical criteria. First, it must reduce TDS on high-demand datasets without harming low-demand controls. Second, it must beat selection-only and low-pass baselines at matched end-to-end latency. Third, its event weights must align with dense evidence maps and human-interpretable key frames.

# 9 Contributions

1. **Problem framing**: stride-induced accuracy loss formalised as pre-decimation evidence loss, grounded in large-scale empirical analysis ($\rho$=+0.26 confidence concentration vs. aliasing loss, 8 datasets).

2. **Eva**: architecture-agnostic dual-band temporal window compression that preserves both low-frequency context and high-frequency event residuals before reducing token rate.

3. **Evidence-supervised training**: dense-to-sparse distillation, explicit evidence-map alignment, and phase-consistency losses targeting the failure modes revealed by TDS.

4. **Backbone-agnostic SOTA evaluation**: Eva is tested across transformer, SSM, and Mamba-3 backbones against adaptive sampling, low-pass anti-aliasing, token compression, and end-to-end latency baselines.

Table 1: Main table template for FineGym/AUTSL/SSv2. Report end-to-end latency, not backbone-only latency.

| Method | $s$=1 | $s$=8 | $s$=16 | TDS↓ | ms↓ |
|---|---|---|---|---|---|
| Uniform stride | – | – | – | – | – |
| LPF + stride | – | – | – | – | – |
| Adaptive selector | – | – | – | – | – |
| VideoMamba | – | – | – | – | – |
| Eva+VideoMamba | – | – | – | – | – |
| Eva+TimeSformer | – | – | – | – | – |
| **Eva-TimeSformer** | – | – | – | – | – |

# References


[1] D. Shao, Y. Zhao, B. Dai, and C. C. Loy, "FineGym: A hierarchical video dataset for fine-grained action understanding," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020.

[2] O. M. Sincan and H. Y. Keles, "AUTSL: A large scale multi-modal Turkish sign language dataset and baseline methods," *IEEE Access*, 2020.

[3] R. Goyal, S. Ebrahimi Kahou, V. Michalski, J. Materzyńska, S. Westphal, H. Kim, V. Haenel, I. Fruend, P. Yianilos, M. Mueller-Freitag, *et al.*, "The "something something" video database for learning and evaluating visual common sense," in *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2017.

[4] K. Soomro, A. R. Zamir, and M. Shah, "UCF101: A dataset of 101 human actions classes from videos in the wild," in *arXiv preprint arXiv:1212.0402*, 2012.

[5] H. Kuehne, H. Jhuang, E. Garrote, T. Poggio, and T. Serre, "HMDB: A large video database for human motion recognition," in *IEEE/CVF*

*International Conference on Computer Vision (ICCV)*, 2011.

[6] Anonymous, "Spatiotemporal aliasing in video action recognition: A cross-architecture analysis at scale," tech. rep., Anonymous, 2026. Under review.

[7] D. Tran, L. Bourdev, R. Fergus, L. Torresani, and M. Paluri, "Learning spatiotemporal features with 3D convolutional networks," in *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2015.

[8] C. Feichtenhofer, H. Fan, J. Malik, and K. He, "SlowFast networks for video recognition," in *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019.

[9] G. Bertasius, H. Wang, and L. Torresani, "Is space-time attention all you need for video understanding?," in *International Conference on Machine Learning (ICML)*, 2021.

[10] A. Arnab, M. Dehghani, G. Heigold, C. Sun, M. Lučić, and C. Schmid, "ViViT: A video vision transformer," in *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021.

[11] Z. Tong, Y. Song, J. Wang, and L. Wang, "VideoMAE: Masked autoencoders are data-efficient learners for self-supervised video pre-training," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.

[12] K. Li, X. Li, Y. Wang, Y. He, Y. Wang, L. Wang, and Y. Qiao, "VideoMamba: State space model for efficient video understanding," in *European Conference on Computer Vision (ECCV)*, 2024. arXiv:2403.06977.

[13] D.-A. Huang, V. Ramanathan, D. Mahajan, L. Torresani, M. Paluri, L. Fei-Fei, and J. C. Niebles, "What makes a video a video: Analyzing temporal information in video understanding models and datasets," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018.

[14] L. Sevilla-Lara, S. Zha, Z. Yan, V. Goswami, M. Feiszli, and L. Torresani, "Only time can tell: Discovering temporal data for temporal modeling," in *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2021.

[15] L. Wang, Y. Xiong, Z. Wang, Y. Qiao, D. Lin, X. Tang, and L. Van Gool, "Temporal segment networks: Towards good practices for deep action recognition," in *European Conference on Computer Vision (ECCV)*, 2016.

[16] J. Lin, C. Gan, and S. Han, "TSM: Temporal shift module for efficient video understanding," in *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019.

[17] Z. Wu, C. Xiong, C.-Y. Ma, R. Socher, and L. S. Davis, "AdaFrame: Adaptive frame selection for fast video recognition," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019.

[18] Y. Zhi, Z. Tong, L. Wang, and G. Wu, "MGSampler: An explainable sampling strategy for video action recognition," in *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021.

[19] S. N. Gowda, M. Rohrbach, and L. Sevilla-Lara, "SMART frame selection for action recognition," in *AAAI Conference on Artificial Intelligence*, 2021.

[20] A. Ghodrati, B. E. Bejnordi, and A. Habibian, "FrameExit: Conditional early exiting for efficient video recognition," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021.

[21] Y. Wang, Z. Chen, H. Jiang, S. Song, Y. Han, and G. Huang, "AdaFocus: Adaptive focus for efficient video recognition," in *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021.

[22] Y. Yang, Y. Wu, Y. Li, G. Li, and L. Lin, "MAFormer: A transformer network with multi-scale attention fusion for visual recognition," in *arXiv preprint arXiv:2209.01620*, 2022.

[23] R. Zhang, "Making convolutional networks shift-invariant again," in *International Conference on Machine Learning (ICML)*, 2019.

[24] C. Jin, Y. Chen, G. Li, T. Zhang, and T. Li, "Low pass filter for anti-aliasing in temporal action localization," *arXiv preprint arXiv:2104.11403*, 2021.

[25] A. Gu, K. Goel, and C. Ré, "Efficiently modeling long sequences with structured state spaces," in *International Conference on Learning Representations (ICLR)*, 2022.

[26] A. Gu, A. Gupta, K. Goel, and C. Ré, "On the parameterization and initialization of diagonal state space models," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.

[27] A. Gu and T. Dao, "Mamba: Linear-time sequence modeling with selective state spaces," in *arXiv preprint arXiv:2312.00752*, 2023.

[28] T. Dao and A. Gu, "Transformers are SSMs: Generalized models and efficient algorithms through structured state space duality," in *International Conference on Machine Learning (ICML)*, 2024.

[29] A. Lahoti, K. Y. Li, B. Chen, C. Wang, A. Bick, J. Z. Kolter, T. Dao, and A. Gu, "Mamba-3: Improved sequence modeling using state space principles," in *International Conference on Learning Representations (ICLR)*, 2026. Oral presentation. arXiv:2603.15569.

[30] Y. Li, Y. Li, and N. Vasconcelos, "Fine-grained action retrieval through multiple parts-of-speech embeddings," in *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019.

[31] D. Bolya, C.-Y. Fu, X. Dai, P. Zhang, C. Feichtenhofer, and J. Hoffman, "Token merging: Your ViT but faster," in *International Conference on Learning Representations (ICLR)*, 2023.

[32] M. Dutson, J. Robinson, B. Bashirzadeh, and M. Riegler, "Eventful transformers: Leveraging temporal redundancy in vision transformers," in *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023.